

Course Code	Name of the Course			
216U03C404	Signals and Systems			
Teaching Scheme	TH	P	TUT	Total
	03	-	01	04
Credits Assigned	03	-	01	04
Evaluation Scheme	Marks			
	LAB/TUT CA	CA (TH)		ESE
		IA	ISE	
	25	20	30	50
				125

Course pre-requisites:

- Basic calculus, Applied Mathematics-I, Applied Mathematics-II

Course Objective:

This course aims to introduce concept of signals and systems and their analysis both in continuous time and discrete time, in preparation for more advanced subjects in digital signal processing including audio, image and video processing, communication theory.

Course Outcomes (CO):

At the end of successful completion of the course students will be able to

- CO1.** Classify signals and perform basic operations on continuous and discrete time signals.
- CO2.** Classify systems and analyse continuous and discrete time Linear Time Invariant system in time domain.
- CO3.** Analyse discrete time Linear Time Invariant system using Z-Transform.
- CO4.** Represent signals and understand properties of CT systems in frequency domain
- CO5.** Represent signals and understand properties of DT systems in frequency domain.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	Basics of Signals and Systems		10	CO1
	1.1	Introduction to Signals and system Explanation of a broad applicability of signals & system applied to mechanical systems, electrical systems, financial systems and biological systems.		
	1.2	Classification of signals: Continuous (CT) and Discrete time (DT) signals, Deterministic and Random Signals, Periodic and non-periodic, even and odd, energy and power signals, Causal and Non-casual signals.		
	1.3	Elementary DT and CT signals Unit Step signal, Unit Ramp signal, Unit impulse, exponential and sinusoidal signals. Operations on signals		
2	Continuous Time and Discrete Time Systems		10	CO2
	2.1	Properties of DT and CT systems: Linearity, causality, time variance, stability and static/dynamic		
	2.2	Discrete Time LTI system: Representation of discrete time signals in terms of impulses, Unit impulse response and convolution sum. Block diagram and difference equation representation of DT-systems (Solution of difference equations using classical method is not expected)		
	2.3	Continuous Time LTI system: Representation of continuous time signals in terms of impulses, Unit impulse response and convolution integral, Block diagram and differential equation representation of CT-systems (Solution of differential equations using classical method is not expected)		
3	Z-Transform		8	CO3
	3.1	Definition, Properties of Z-Transform (No derivations), Z-Transforms of standard signals, unilateral Z-Transform, Inverse Z-Transform.		
	3.2	Analysis and Characterization of DT-LTI Systems using Z-Transform: Relation of Unit Impulse Response and System Function, Response of DT system using Z-transform, properties of DT-LTI system in Z-domain.		
Self-study: derivations of important properties of Z-Transform, Concept map for DT-LTI system; Relation: DT system-Block diagram, difference equation, unit impulse response, system function				

4	Frequency characterization of CT signals and LTI systems		9	CO4
	4.1	CT Fourier Series (CTFS): Response of LTI systems to complex exponentials, Fourier series: Representing periodic signals in terms of Sinusoids and harmonics, Orthogonality, Trigonometric and Exponential of Fourier series, properties of CTFS (No derivations), filters described by differential equation.		
	4.2	CT Fourier Transform (CTFT): Representing CT aperiodic signals by Fourier Transform – Definition of CTFT, CTFT of CT aperiodic signals, Properties of CTFT (No derivations). Inverse Fourier Transform, Magnitude and Phase response of CT-LTI systems.		
5	Frequency characterization of DT Signals and LTI systems		8	CO5
	5.1	DT Fourier Series (DTFS): Fourier Series representation of DT periodic Signals, properties of DTFS (No derivations), filters described by difference equations		
	5.2	DT Fourier Transform (DTFT): Representing DT aperiodic signals by DT Fourier Transform – Definition of DTFT, Properties of DTFT (No derivations), Inverse DTFT, Magnitude and Phase response of DT-LTI systems.		
Self-study: Derivations of important properties of Fourier Transform, Concept map showing relations among CTFS and DTFS, CTFT and DTFT				
Total			45	

Application areas: Speech signal and model of speech production system, Phonograph, Electrical circuits using resistors, capacitors and operational amplifiers, Hydraulic systems, Mass & Spring suspension system, R-C filters (low pass and high pass – first and second order), Communication systems (e.g. cell phone system), Computer system (e.g. accumulator)

Reference Books:

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Alan Oppenheim, Alan S. Willsky, Syed H. Nawab	Signals and Systems	Prentice Hall	Second/2015
2	Alan V Oppenheim and S Hamid	Signal and System	Person New International Edition	Second/ January 2015
3	Kumar A. Anand	Signals and Systems	Prentice Hall India	Third /2013
4	Simon Haykin	Signals and Systems	Wiley	Second/ 2007
5	Tarun Rawat	Signals and Systems	Oxford Higher Education	First/2010
6	Alberto Leon-Garcia	Probability and Random Processes for Electrical Engineering	Pearson, UKcountry of origin	Second edition, 1993

LAB/TUT CA:	Tutorials includes journal / written record of tutorials. Tutorials will be evaluated out of 25 marks each . A minimum of 8 tutorials to be performed in a semester based on complete syllabus .
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